

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Operating Systems | Course Code: 21CS44 | Credits: 3

Exam Session: End Semester Exam 2023

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

- Q1.** Consider 5 processes P0 to P4 with Allocation, Max matrices, and Available resources [3, 3, 2] for resources (A, B, C). Compute the Need Matrix and verify if the system is in a Safe State using Banker's Algorithm. List the safe sequence. [8 Marks]

--- OR ---

- Q2.** For the reference string: [7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1] with 3 physical frames, compute the number of page faults using: (i) FIFO, (ii) LRU, and (iii) Optimal Page Replacement. [8 Marks]